Hugo Marquerie 13/02/2025

Teorema de Picard-Lindelöf

Teorema 1 (de Picard-Lindelöf). *Sea* $f\colon \Omega \subset \mathbb{R} \times \mathbb{R}^d \longrightarrow \mathbb{R}^d$ *una función continua y localmente Lipschitz en la segunda variable uniforme en la primera*

$$\implies \forall (t_0, \overline{x}) \in \Omega : \exists \varepsilon > 0 : \exists!\, x \in \mathcal{C}^1([t_0{-}\varepsilon, t_0{+}\varepsilon]) : \begin{cases} \forall t \in [t_0 - \varepsilon, t_0 + \varepsilon] : x'(t) = f(t, x(t)) \\ x(t_0) = \overline{x} \end{cases}$$

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